
Experiment 2

Creating a Single Line Diagram in PSS®E

Lab Report

Student Name HAMZA AHMAD

Registration Number 2022-EE-165

Section B

Power System Operation and Control Laboratory

Fall 2025

The load flow study using the Newton-Raphson method converged successfully, indicating that the system data and parameters were correctly entered. Convergence was achieved within the standard tolerance limits, ensuring reliable and stable power flow solutions for the IEEE-14 bus system.

Bus Voltage Profile

The bus voltage magnitudes across the system were within acceptable limits, generally close to the nominal base voltage (1.0 p.u.). The slack bus (Bus 1) voltage was fixed at 1.06 p.u., serving as the voltage reference point. PV buses (Buses 2, 3, 6, and 8) maintained voltage setpoints through generator control, while PQ buses showed voltage variations based on load demands. No bus voltage violations (e.g., below 0.95 p.u. or above 1.05 p.u.) were observed, indicating good voltage regulation in the network.

Power Flows in Lines and Transformers

Power flows on transmission lines and transformers were consistent with the IEEE-14 bus system model. The flow direction and magnitude aligned with the expected power transfer from generation to load buses. All line loadings were within thermal limits, confirming that no branch was overloaded during the steady-state operation simulated.

Generator Outputs

The generators at buses 1, 2, 3, 6, and 8 supplied active and reactive power as per system requirements, balancing total generation with load plus losses. The slack generator at Bus 1 adjusted its output to compensate for system losses and maintain the system reference voltage angle.

Comparison with Standard IEEE-14 Bus Results

The obtained bus voltages, line flows, and generator outputs closely matched standard IEEE-14 bus load flow results reported in literature and PSS®E example cases (ieee_gic_test_case.sav). This validates the correctness of the case data input and load flow solution procedure.

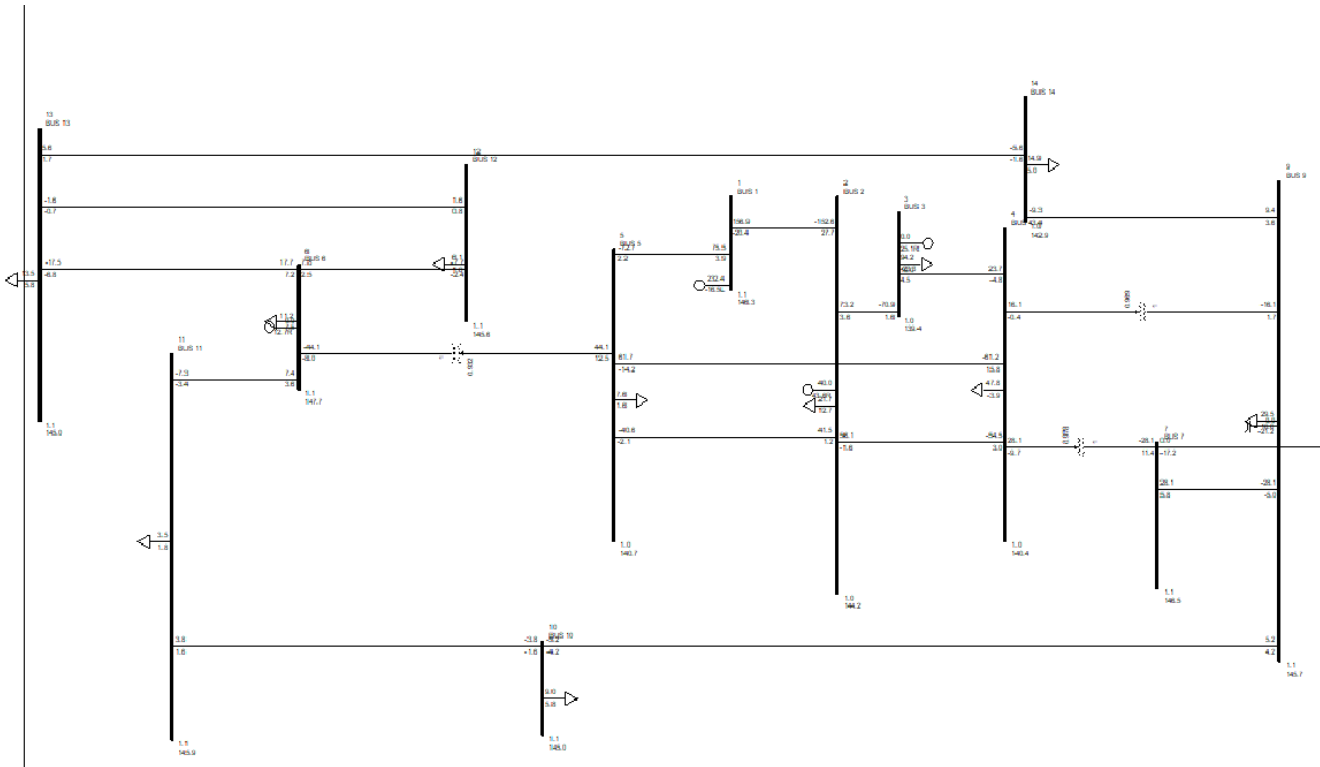
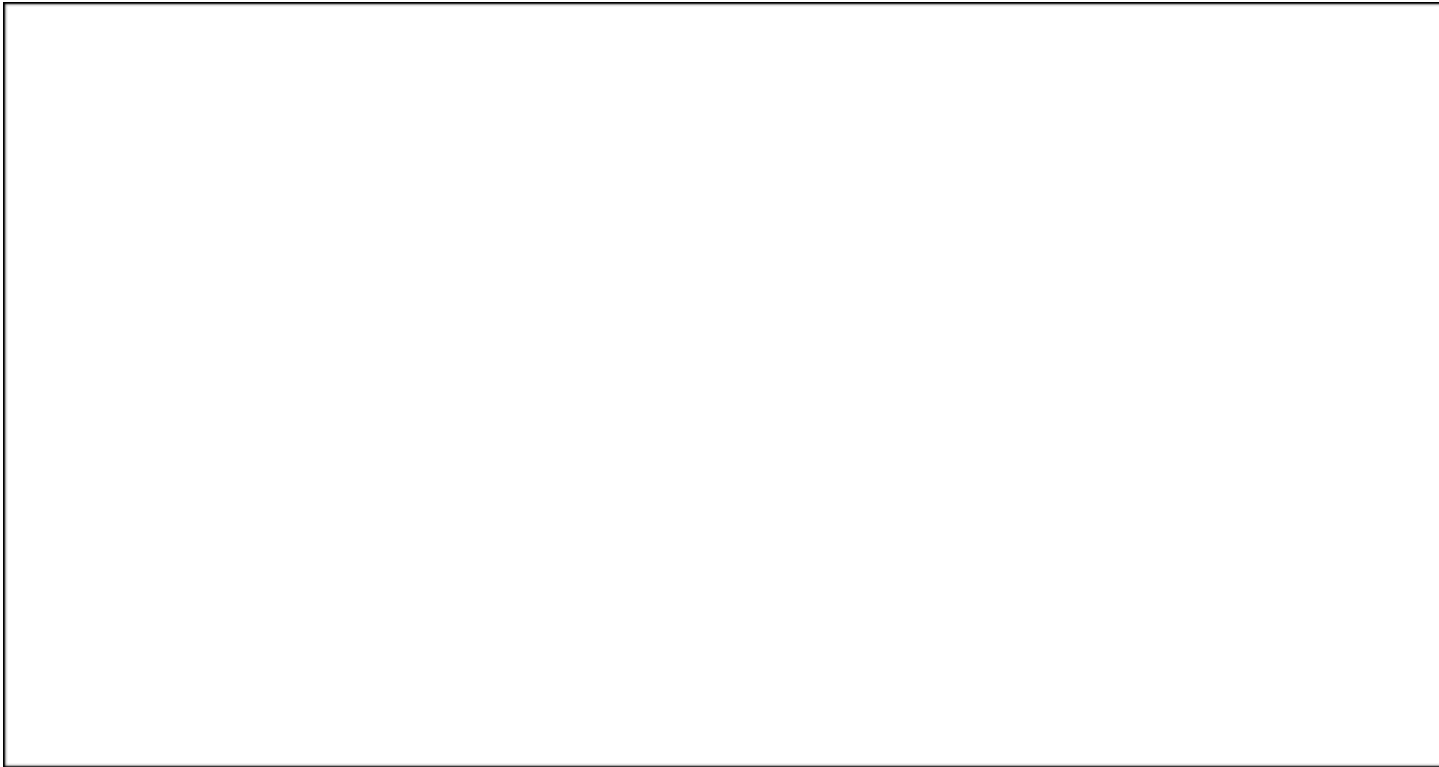
Observations and Challenges

Initial data entry required careful attention to bus types and branch parameters to ensure convergence.

The graphical SLD created in the PSS®E editor provided a clear visualization of system topology and aided in cross-verifying the connectivity of components.

Minor adjustments in bus placement and labeling improved diagram clarity and result interpretation.

Screenshot of final SLDs.



Screenshots of results (bus voltages)



IEEE14.SLD		Network data ✕													
	Bus Number	Bus Name	Base kV	Area	Area Name	Zone	Zone Name	Owner	Owner Name	Code	Voltage (pu)	Angle (deg)	Normal Vmax (pu)	Normal Vmin (pu)	Emergency Vmax (pu)
	1	BUS 1	138.0	1	IEEE14	1	IEEE 14	1	1	3	1.0600	0.00	1.1000	0.9000	1.1
	2	BUS 2	138.0	1	IEEE14	1	IEEE 14	1	1	2	1.0450	-4.98	1.1000	0.9000	1.1
	3	BUS 3	138.0	1	IEEE14	1	IEEE 14	1	1	2	1.0100	-12.73	1.1000	0.9000	1.1
	4	BUS 4	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0177	-10.31	1.1000	0.9000	1.1
	5	BUS 5	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0195	-8.77	1.1000	0.9000	1.1
	6	BUS 6	138.0	1	IEEE14	1	IEEE 14	1	1	2	1.0700	-14.22	1.1000	0.9000	1.1
	7	BUS 7	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0615	-13.36	1.1000	0.9000	1.1
	8	BUS 8	138.0	1	IEEE14	1	IEEE 14	1	1	2	1.0900	-13.36	1.1000	0.9000	1.1
	9	BUS 9	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0559	-14.94	1.1000	0.9000	1.1
	10	BUS 10	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0510	-15.10	1.1000	0.9000	1.1
	11	BUS 11	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0569	-14.79	1.1000	0.9000	1.1
	12	BUS 12	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0552	-15.08	1.1000	0.9000	1.1
	13	BUS 13	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0504	-15.16	1.1000	0.9000	1.1
	14	BUS 14	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0355	-16.03	1.1000	0.9000	1.1
*															

Screenshots of results (line flows, generator outputs)

IEEE14.SLD		Network data															
	From Bus	From Bus Name	To Bus Number	To Bus Name	Id	Line R (pu)	Line X (pu)	Charging B (pu)	In Service	Metered	Rate A	Rate B	Rate C	Line G From (pu)	Line B From (pu)	Line G To (pu)	Line B To (pu)
	1	BUS 1	1	2 BUS 2	1	0.019380	0.059170	0.052800	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	1	BUS 1	1	5 BUS 5	1	0.054030	0.223040	0.049200	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	2	BUS 2	1	3 BUS 3	1	0.046990	0.197970	0.043800	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	2	BUS 2	1	4 BUS 4	1	0.058110	0.176320	0.034000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	2	BUS 2	1	5 BUS 5	1	0.056950	0.173880	0.034600	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	3	BUS 3	1	4 BUS 4	1	0.067010	0.171030	0.012800	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	4	BUS 4	1	5 BUS 5	1	0.013350	0.042110	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	6	BUS 6	1	11 BUS 11	1	0.094980	0.198900	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	6	BUS 6	1	12 BUS 12	1	0.122910	0.255810	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	6	BUS 6	1	13 BUS 13	1	0.066150	0.130270	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	7	BUS 7	1	8 BUS 8	1	0.000000	0.176150	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	7	BUS 7	1	9 BUS 9	1	0.000000	0.110010	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	9	BUS 9	1	10 BUS 10	1	0.031810	0.084500	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	9	BUS 9	1	14 BUS 14	1	0.127110	0.270380	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	10	BUS 10		11 BUS 11	1	0.082050	0.192070	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	12	BUS 12		13 BUS 13	1	0.220920	0.199880	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
	13	BUS 13		14 BUS 14	1	0.170930	0.348020	0.000000	☒	☒ Fro	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.00000
*									☒	☒ Fro							

IEEE14.SLD		Network data ✕																
	Bus Number	Bus Name	Id	Area	Area Name	Zone	Zone Name	Cod	VSched (pu)	Remote Bus	In Serv	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)	Mbase (MVA)
	1	BUS 1	1	1	IEEE14	1	IEEE 14	3	1.0600	0	☒	232.3930	10000.0000	10000.0000	-16.5495	0.0000	0.0000	615.00
	2	BUS 2	1	1	IEEE14	1	IEEE 14	2	1.0450	0	☒	40.0000	10000.0000	10000.0000	43.5572	50.0000	-40.0000	60.00
	3	BUS 3	1	1	IEEE14	1	IEEE 14	2	1.0100	0	☒	0.0000	10000.0000	10000.0000	25.0754	40.0000	0.0000	60.00
	6	BUS 6	1	1	IEEE14	1	IEEE 14	2	1.0700	0	☒	0.0000	10000.0000	10000.0000	12.7309	24.0000	-6.0000	25.00
	8	BUS 8	1	1	IEEE14	1	IEEE 14	2	1.0900	0	☒	0.0000	10000.0000	10000.0000	17.6235	24.0000	-6.0000	25.00
*											☒							